

Q1. What is machine learning?

A. Machine learning is a subset of AI that enables systems to learn patterns from data and improve performance without explicit programming.

Q2. AI vs ML?

A. AI is the broader concept of intelligent machines. ML is a subset of AI that learns from data.

Q3. Deep learning?

A. Deep learning is a branch of ML that uses multi-layer neural networks to learn complex patterns.

Q4. Neural networks?

A. Neural networks are interconnected layers of artificial neurons that process and learn from data.

Q5. Backpropagation?

A. Backpropagation updates neural network weights by propagating prediction errors backward through the network.

Q6. Gradient descent?

A. Gradient descent is an optimization algorithm that minimizes loss by adjusting model parameters iteratively.

Q7. Activation functions?

A. Activation functions introduce non-linearity into neural networks, enabling them to learn complex relationships.

Q8. What is ReLU?

A. ReLU (Rectified Linear Unit) outputs $\max(0, x)$ and is widely used because it is simple and efficient.

Q9. What is sigmoid?

A. The sigmoid function maps values between 0 and 1 and is often used for binary classification outputs.

Q10. CNNs?

A. Convolutional Neural Networks are specialized for image and spatial data processing.

Q11. RNNs?

A. Recurrent Neural Networks process sequential data by maintaining information from previous inputs.

Q12. LSTMs?

A. Long Short-Term Memory networks are RNN variants designed to handle long-term dependencies.

Q13. Transformers?

A. Transformers use self-attention mechanisms to process sequences efficiently and power modern LLMs.

Q14. Attention mechanism?

A. Attention allows models to focus on the most relevant parts of input data.

Q15. What is BERT?

A. BERT is a bidirectional transformer model designed for language understanding tasks.

Q16. Explain GPT models

A. GPT models are transformer-based language models trained to predict the next token in a sequence.

Q17. Transfer learning?

A. Transfer learning reuses knowledge from a pre-trained model for a new task.

Q18. Fine-tuning?

A. Fine-tuning adapts a pre-trained model to a specific dataset or task.

Q19. What is RAG?

A. Retrieval-Augmented Generation combines information retrieval with LLM generation for more accurate responses.

Q20. Embeddings?

A. Embeddings are dense vector representations that capture semantic meaning of data.

Q21. Vector databases?

A. Vector databases store and search embeddings efficiently for semantic retrieval.

Q22. Prompt engineering?

A. Prompt engineering is the practice of designing prompts to improve AI outputs.

Q23. Hallucinations?

A. Hallucinations occur when AI generates incorrect or fabricated information.

Q24. Reinforcement learning?

A. Reinforcement learning trains agents through rewards and penalties from interactions with an environment.

Q25. What is PPO?

A. Proximal Policy Optimization is a reinforcement learning algorithm that stabilizes policy updates.

Q26. What is RLHF?

A. Reinforcement Learning from Human Feedback aligns model behavior with human preferences.

Q27. Overfitting?

A. Overfitting occurs when a model memorizes training data and performs poorly on unseen data.

Q28. Regularization?

A. Regularization techniques reduce overfitting by constraining model complexity.

Q29. Dropout?

A. Dropout randomly disables neurons during training to improve generalization.

Q30. Batch normalization?

A. Batch normalization normalizes layer inputs to improve training speed and stability.

Q31. Model quantization?

A. Quantization reduces model size by using lower-precision numerical representations.

Q32. Pruning?

A. Pruning removes unnecessary model parameters to improve efficiency.

Q33. Inference?

A. Inference is the process of using a trained model to make predictions.

Q34. Training pipelines?

A. Training pipelines automate data preparation, model training, evaluation, and deployment steps.

Q35. What is MLOps?

A. MLOps applies DevOps practices to machine learning lifecycle management.

Q36. Model serving?

A. Model serving makes trained models available for real-time or batch predictions.

Q37. TensorFlow?

A. TensorFlow is an open-source machine learning framework developed by Google.

Q38. PyTorch?

A. PyTorch is an open-source deep learning framework known for flexibility and ease of use.

Q39. Hugging Face?

A. Hugging Face provides open-source tools, datasets, and pre-trained AI models.

Q40. LangChain?

A. LangChain is a framework for building LLM-powered applications and workflows.

Q41. Vector search?

A. Vector search finds similar items based on embedding distance rather than exact keywords.

Q42. Cosine similarity?

A. Cosine similarity measures the angle between vectors to determine similarity.

Q43. Evaluation metrics?

A. Metrics such as accuracy, precision, recall, F1-score, and ROC-AUC evaluate model performance.

Q44. Confusion matrix?

A. A confusion matrix summarizes classification results using TP, TN, FP, and FN values.

Q45. Precision-recall?

A. Precision measures correctness of positive predictions; recall measures how many actual positives are found.

Q46. Describe an ML project

A. Explain the problem, data collection, preprocessing, model selection, evaluation, deployment, and outcomes.

Q47. Deploy an LLM

A. Deploy using model serving infrastructure, APIs, monitoring, scaling, and security controls.

Q48. Reduce hallucinations

A. Use RAG, better prompts, fine-tuning, validation, and high-quality data sources.

Q49. Evaluate AI systems

A. Assess accuracy, robustness, latency, fairness, safety, and business impact.

Q50. Questions for us?

A. Ask about team structure, success metrics, challenges, growth opportunities, and technology stack.